

Sets, Events & Bayesian Inference

From set notation to updating beliefs with evidence

How to Use This Handout

This document covers one continuous arc: from the basic notation for describing events, through the rules that govern how probabilities combine, and finally to Bayes' theorem and Bayesian inference.

Each idea depends on the one before it. Bayes' theorem is not a standalone trick. It is what you get when you take conditional probability and the multiplication rule and rearrange them. So resist the urge to skip ahead. The last section will feel obvious if you build up to it, and impossible if you don't.

The three parts:

- **Part I — Sets and Events.** How to describe events, draw them, and combine them.
- **Part II — The Rules of Probability.** How probabilities behave when events interact.
- **Part III — Bayes and Bayesian Inference.** How to update what you believe when evidence arrives.

Practice problems and worked answers appear at the end of each part.

PART I — SETS AND EVENTS

1. Events as Sets

Every event has a set of outcomes that satisfy it. These are the **favorable outcomes**.

For example, if the event is "rolling an even number," the set of values consists of 2, 4, 6, and so on.

The values in a set do not have to be numerical. If the event is "being a member of the European Union," then France and Germany belong to the set, while the USA and Japan do not.

Notation convention

- **Uppercase letters** denote sets. A , B , X
- **Lowercase letters** denote individual elements. a , b , x

So if X is the set of all even numbers, then x could be the single value 8.

Empty and non-empty sets

A set either contains values or it does not.

- If it contains no values at all, we call it the **empty set** or **null set**, written $\emptyset$. The symbol looks like a crossed-out zero.
- Non-empty sets are either **finite** or **infinite**, depending on how many elements they hold.

2. Set Notation

You will see these symbols constantly. Learn them now and the rest of the course reads more easily.

Symbol	Read as	Meaning
$x \in A$	"x is an element of A" or just "x in A"	The value x belongs to set A
$A \ni x$	"A contains x"	Same statement, written from the set's side
$x \notin A$	"x is not in A"	The value x does not belong to A
$A \not\ni x$	"A does not contain x"	Same statement, flipped
$\forall$	"for all" or "for any"	A capital A turned upside down
$:$	"such that"	Narrows a statement to a specific group
$\subseteq$	"is a subset of"	Every element of the first set is in the second

On the "element of" symbol

Having two ways to say the same thing ($x \in A$ and $A \ni x$) may look redundant. It is not. Depending on what you are emphasizing, one reads more naturally than the other, and you will meet both in textbooks.

To negate either one, cross the symbol with a single diagonal line.

On "for all" and "such that"

The $\forall$ symbol lets you make a general statement about every element of a set:

$\forall x \in A$	"for all x in A "
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The colon, read as "such that," lets you narrow that statement to a subgroup:

$\forall x \in A : x \text{ is even}$	"for all x in A , such that x is even"
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This combination is extremely common. Get comfortable reading it out loud.

Subsets

A **subset** is a set fully contained inside another set.

If every element of A is also an element of B , then A is a subset of B . We write $A \subseteq B$.

Note that the reverse need not hold. Not all elements of B are necessarily in A .

Remember this rule: every set contains at least two subsets, itself and the null set.

3. Visualizing Events

We draw the set of values satisfying an event as a **circle**. One circle per event. Any element belonging to a set appears as a point inside the appropriate circle. More events means more circles.

Take two events, A and B . There are exactly **three** possible relationships between their circles.

Case 1: The circles never touch

The two events can never happen at the same time. Event A occurring guarantees event B is not occurring, and the reverse.

Example: drawing a diamond and drawing a heart. Every card has exactly one suit, so a heart cannot also be a diamond.

This is the simplest case, and it is the one assumed in most introductory examples.

Case 2: The circles intersect

The two events **can** occur at the same time.

Example: draw one card. Let A be "drawing a diamond" and B be "drawing a queen."

- The overlapping area contains exactly one card: the queen of diamonds.
- The rest of A holds all other diamonds.
- The rest of B holds all other queens.

Case 3: One circle completely contains the other

One event can only ever occur if the other one does as well.

Example: A is "drawing a red card" and B is "drawing a diamond." Circle B sits entirely inside circle A.

Read the logic carefully, because this is where students slip:

- If the card is black, event A did not occur, and therefore event B cannot have occurred either.
- But you can draw a heart. So a red card that is not a diamond is perfectly possible.
- Therefore **event B not occurring does not guarantee event A did not occur.**

In short: if an outcome is not part of a set, it cannot be part of any of its subsets. But an outcome not being part of a subset does not exclude it from the greater set.

With more than two events

You can have three, four, or as many events as you like. The relationship between any **pair** of them will always be one of these three cases. Nothing new appears.

4. Intersections

When we want **both A and B to happen at the same time**, we are talking about their **intersection**.

Graphically it is exactly what the name suggests: the area where the circles overlap. It consists of all outcomes favorable to A and B simultaneously.

$$A \cap B$$

The three cases, revisited

Circles do not touch: the intersection is the empty set. There is no outcome satisfying both.

$$A \cap B = \emptyset$$

All hearts and all diamonds. No card is both.

Circles intersect: the intersection holds whatever they share.

Diamonds and queens intersect at the queen of diamonds. That single card is the only one satisfying both conditions, so nothing else is included.

One circle inside the other: the intersection is the smaller set.

$$A \cap B = B$$

Red cards and diamonds. Every diamond is simultaneously red and a diamond, so the intersection is simply all diamonds.

Use intersections only when you need both events to happen at once.

5. Unions

When we only require **one of them to occur**, regardless which, we are talking about the **union**. This is the same as asking for either A or B.

$$A \cup B$$

The symbol is the intersection symbol flipped upside down. To avoid mixing them up, remember that the union symbol resembles a capital letter **U**.

An intuition

Think of American states as separate sets, with citizens as their elements. Just as every Texan is an American, every element of an individual set is also an element of the union.

The three cases

Circles do not touch: the union is simply the sum of the two sets.

Their intersection is empty, so there is no risk of double-counting. Hearts and diamonds together give you all red cards. No card has two suits, so no card gets counted twice.

Circles intersect: the union is the sum of the two sets **minus their intersection**.

If you simply added the two areas, every element inside the overlap would be counted twice.

Worked example. In an office, 5 people have blonde hair and 4 people have blue eyes. Only Kate has both. That leaves 3 non-blonde people with blue eyes and 4 blonde people without blue eyes.
Counting directly: $3 + 4 + 1 = 8$ people with at least one of the two features.
Using the formula: $5 + 4 - 1 = 8$. The minus one is Kate, who sits in the intersection.

One circle inside the other: the union is simply the larger set.

Let A be "being from the U.S." and B be "being from California." Everyone who is either Californian or American is just... everyone American.

The formula confirms it. The intersection of A and B is all Californians, so:

$$\begin{aligned} A \cup B &= A + B - (A \cap B) \\ &= \text{all Americans} + \text{all Californians} - \text{all Californians} \\ &= \text{all Americans} \end{aligned}$$

The general formula

$$A \cup B = A + B - (A \cap B)$$

This is universally true, regardless of the relationship between A and B. In the non-touching case the intersection is zero, so it reduces to a plain sum. In the subset case the intersection cancels the smaller set out. One formula, all three situations.

6. Mutually Exclusive Sets and Complements

Mutually exclusive sets

Mutually exclusive sets are sets that are not allowed to have any overlapping elements. Graphically, their circles never intersect.

Two properties follow directly:

1. **Their intersection is the empty set.** This works in both directions. If the intersection of any number of sets is empty, they are mutually exclusive, and if they are mutually exclusive, their intersection is empty.
2. **Their union is simply the sum of the individual sets.** No subtraction needed, because there is nothing to double-count.

Complements

Just as events have complements, so do sets. The **complement** of a set consists of all values that are part of the sample space but not part of the set.

Example: if a set contains all the odd numbers, its complement is the set of all even numbers.

The distinction that trips people up

Complements are always mutually exclusive. But not all mutually exclusive sets are complements.

Counterexample. Let A be the set of all even numbers, and B the set of all numbers ending in five.

- Any number ending in five is odd, so A and B are definitely mutually exclusive.
- But the complement of "all even" is "all odd," not just the odd numbers ending in five.
- The number 13 is in the complement of A, but it is not in B.

So A and B are mutually exclusive without being complements.

Practice — Part I

Problem 1. From a standard 52-card deck, let A be "drawing a face card" (J, Q, K) and B be "drawing a club." How many cards are in $A \cap B$? How many in $A \cup B$?

Problem 2. Roll a single die. Let A be "rolling an even number" and B be "rolling a prime number." Are A and B mutually exclusive? What is the complement of A?

Problem 3. In an office of 30 people, 18 drink coffee, 14 drink tea, and 6 drink both. How many drink at least one of the two? How many drink neither?

Answers — Part I

Problem 1. A has 12 elements (3 face cards $\times$ 4 suits). B has 13. $A \cap B$ = the jack, queen, and king of clubs = **3 cards**.
 $A \cup B$ = $12 + 13 - 3 =$ **22 cards**.

Problem 2. $A = \{2, 4, 6\}$. $B = \{2, 3, 5\}$. Their intersection is $\{2\}$, which is **not empty**, so they are **not mutually exclusive**. The complement of A is $\{1, 3, 5\}$, the odd numbers.

Problem 3. At least one: $18 + 14 - 6 =$ **26 people**. Neither: $30 - 26 =$ **4 people**.

PART II — THE RULES OF PROBABILITY

7. Independent and Dependent Events

Independent events

Independent events have theoretical probabilities that remain unaffected by other events.

The simplest example is flipping a coin. You always have a 50% chance of tails, regardless of what the previous throw produced.

Dependent events

Dependent events have probabilities that vary as conditions change.

Take the probability of drawing the queen of spades from a full deck. Normally the answer is $1/52$, since there is exactly one favorable outcome and 52 elements in the sample space.

Now watch what happens as information arrives.

What you know	New sample space	Probability
Nothing	52 cards	$1/52$
The card is a spade	13 cards	$1/13$
The card is a queen	4 cards	$1/4$

The probability of an event changes depending on the information we have. This is the central insight of this entire section, and everything that follows is built on it.

Notation

To express the probability of A given that B has occurred:

$$P(A|B)$$

Read as "**P of A given B.**"

Applying it to the card example, with A as "drawing the queen of spades":

- Let B be "drawing a spade." Then $P(A|B) = 1/13$.
- Let C be "drawing a queen." Then $P(A|C) = 1/4$.

We call this the **conditional probability**, and we use it to distinguish dependent from independent events.

8. Computing Conditional Probability

Conditional probability is the likelihood of an event occurring, assuming a different one has already happened.

Testing for independence

Return to the coin. Let A be "getting heads now" and B be "getting heads on the previous flip."

$$P(A) = 0.5 \qquad P(A|B) = 0.5$$

Since they are equal:

$$P(A) = P(A|B) \quad \Leftrightarrow \quad A \text{ and } B \text{ are independent}$$

That is the test. If knowing B changes nothing about A, the events are independent.

Recall also that **for independent events, the probability of their intersection is the product of the individual probabilities.**

Testing for dependence

Back to the queen of spades. $P(A) = 1/52$ but $P(A|B) = 1/13$. Since these differ, **A and B are dependent.**

Likewise $P(A|C) = 1/4 \neq P(A)$, so A and C are dependent too.

The formula

By definition, the conditional probability of A given B is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

This holds only when $P(B) > 0$.

The restriction is logical, not arbitrary. If $P(B) = 0$, then event B never occurs, so "A given B" describes an impossible situation and cannot be interpreted.

Where the formula comes from

Compare it to the favorable-over-total formula you have been using all along. The similarities are not a coincidence.

- **The numerator.** To satisfy a conditional probability, both A and B must occur simultaneously. So the intersection of A and B holds all the favorable outcomes.
- **The denominator.** The conditional probability requires that B occurs. So the sample space shrinks to all the outcomes where B is satisfied.

It is still favorable over total. The conditioning event just redefines what "total" means.

Order matters enormously

$P(A|B)$ is **not** the same as $P(B|A)$, even when the two happen to have equal numeric values.

Example — Hamilton College, class of 2018.

- 5% of students who got an economics degree graduated with honors.
- 5% of all students who graduated with honors completed a concentration in economics.

Identical percentages, completely different meanings.

- The first says 4 of the 80 economics majors graduated with distinction.
- The second says 4 of the 80 students who graduated with high grades were economics majors.

These describe different groups, different denominators, and different facts about the world.

9. The Law of Total Probability

A survey to work with

100 men and women of all ages were asked if they eat meat.

	Vegetarian	Not vegetarian	Total
Women	15	32	47
Men	29	24	53
Total	44	56	100

Let A be "being vegetarian" and B be "being a woman."

$P(A|B) = 15/47 \approx 0.319$ the likelihood a woman is vegetarian
 $P(B|A) = 15/44 \approx 0.341$ the likelihood a vegetarian is a woman

Same numerator of 15. Different denominators. Different questions.

Since $15/44 > 15/47$, it is more likely for a vegetarian to be female than for a woman to be vegetarian. In probability theory, things are rarely as symmetric as they look.

The law

Suppose A is the union of some finitely many events $B_1, B_2, \dots$ that cover the whole sample space. Then:

$$P(A) = \sum P(A|B_i) \times P(B_i)$$

In words: the probability of A is the sum of all the conditional probabilities of A given some B, each multiplied by the probability of that B.

Applying it

The probability of being vegetarian equals the probability of being vegetarian given male, times the probability of being male, plus the probability of being vegetarian given female, times the probability of being female.

$$\begin{aligned} P(\text{vegetarian}) &= (29/53 \times 53/100) + (15/47 \times 47/100) \\ &= (0.547 \times 0.53) + (0.319 \times 0.47) \\ &= 0.29 + 0.15 \\ &= 0.44 \end{aligned}$$

44%. And the table confirms it directly: 44 of 100 respondents were vegetarian. The answer checks out.

This law is what lets you compute an overall probability by breaking a population into groups, handling each group separately, and weighting the results. You will use it constantly, including in every Bayes calculation in Part III.

10. The Additive Law

Recall the union rule from Part I. The additive law states the same thing in probability terms.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The probability of the union of two sets equals the sum of the individual probabilities minus the probability of their intersection.

Applying it to the survey

The union of women and vegetarians:

$$\begin{aligned} P(\text{woman} \cup \text{vegetarian}) &= 0.47 + 0.44 - 0.15 \\ &= 0.76 \end{aligned}$$

Pick a random person from the survey and there is a **76% chance** they are female, vegetarian, or both.

Rearranging it

The formula is more useful than it first appears, because you can solve for whichever term is missing.

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

Worked example. 38% of your colleagues can proficiently use Tableau. 45% are experts in SQL. 66% are good with at least one of the two.

What is the probability someone can do both?

$$P(\text{both}) = 0.38 + 0.45 - 0.66 = 0.17$$

17%, or a probability of 0.17.

Notice that you were never told the intersection directly. The additive law recovered it from the union.

11. The Multiplication Rule

Start from the conditional probability formula and multiply both sides by $P(B)$:

$$P(A \cap B) = P(A|B) \times P(B)$$

This is the **multiplication rule**.

Why it makes sense

Suppose $P(B) = 0.5$ and $P(A|B) = 0.8$.

Event B occurs 50% of the time. Within that 50%, event A also appears 80% of the time. So the likelihood of both occurring together is:

$$0.8 \times 0.5 = 0.4$$

You are taking a fraction of a fraction. That is all the rule says.

Worked example: two cards, no replacement

Draw one card from a standard deck, shuffle **without returning it**, then draw a second.

Question: what is the probability of drawing a spade on the second draw and **not** drawing a spade on the first?

Step 1 — Define the events.

- A = drawing a spade on the second try
- B = not drawing a spade on the first try

Step 2 — Find $P(B)$.

The likelihood of drawing a specific suit is $1/4 = 0.25$. B is the complement of that:

$$P(B) = 1 - 0.25 = 0.75$$

Step 3 — Find $P(A|B)$. Be careful here.

We have assumed no spade was drawn first, so all 13 spades remain. But one card is gone from the deck, so the sample space is now 51, not 52.

$$P(A|B) = 13/51 \approx 0.255$$

Step 4 — Apply the multiplication rule.

$$P(A \cap B) = 0.255 \times 0.75 \approx 0.191$$

About 19.1%.

Read the answer carefully. 0.191 is the probability of **both** things happening: no spade first and a spade second. It is not the conditional probability, which is 0.255. Students routinely confuse the two. The conditional assumes the first draw already happened. The intersection accounts for the chance that it happened at all.

Practice — Part II

Problem 4. $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \cup B) = 0.65$. Find $P(A \cap B)$. Are A and B independent?

Problem 5. Draw two cards from a full deck without replacement. What is the probability both are hearts?

Problem 6. Using the survey table from Section 9, compute $P(\text{man} \mid \text{vegetarian})$. Compare it to $P(\text{vegetarian} \mid \text{man})$ and explain the difference in plain words.

Answers — Part II

Problem 4.

$$P(A \cap B) = 0.5 + 0.3 - 0.65 = 0.15$$

Independence test: if independent, the intersection should equal the product of the individual probabilities.

$$0.5 \times 0.3 = 0.15 \quad \checkmark$$

They match, so **A and B are independent.**

Problem 5.

$$\begin{aligned} P(\text{first heart}) &= 13/52 = 0.25 \\ P(\text{second heart} \mid \text{first heart}) &= 12/51 \approx 0.235 \\ P(\text{both}) &= 0.25 \times 0.235 \approx 0.059 \end{aligned}$$

About 5.9%. Note that both the favorable count and the sample space dropped by one.

Problem 6.

$$\begin{aligned} P(\text{man} \mid \text{vegetarian}) &= 29/44 \approx 0.659 \\ P(\text{vegetarian} \mid \text{man}) &= 29/53 \approx 0.547 \end{aligned}$$

The first says: of the vegetarians in the survey, about 66% are men. The second says: of the men in the survey, about 55% are vegetarian. Same 29 people, but measured against two different groups, so they answer two different questions.

PART III — BAYES AND BAYESIAN INFERENCE

12. Bayes' Theorem

Also called **Bayes' rule** or **Bayes' law**. The three names are used interchangeably.

Deriving it

Take two events, A and B. Start with the conditional probability formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Now use the multiplication rule to rewrite the numerator. Note that we write it the *other* way around, conditioning on A instead of B:

$$P(A \cap B) = P(B|A) \times P(A)$$

Substitute that in:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

That is Bayes' theorem. It required nothing you did not already have. Two formulas from Part II, combined.

Why it matters

Bayes' theorem lets you find the relationship between the two different conditional probabilities of a pair of events.

That is a bigger deal than it sounds. Very often one direction is easy to measure and the other is the one you actually care about. A lab can easily measure "how often does the test fire in patients who are sick." What a doctor needs is "how likely is this patient sick given the test fired." Bayes converts between them.

Example: medical correlation

There is a correlation between patients with back problems and patients wearing glasses.

- **67%** of people with spinal problems wear glasses.
- Only **41%** of patients with eyesight issues have back pain.

These conditional probabilities suggest the link runs much more strongly in one direction than the other.

We cannot establish a direct causal link, but we can construct arguments. Most patients with back pain are either elderly or work desk jobs where they stay stationary for long periods, and both old age and long hours at a screen can degrade eyesight. Meanwhile, plenty of healthy young people wear glasses from childhood with no underlying factor pointing toward future back pain.

Knowing both conditional probabilities lets you make more reasonable arguments about which factor might drive the other. It does not settle the question, but it sharpens it.

Example: hiring research

Your boss asks what firms should prioritize in recent graduates: good academic performance or working experience?

You review 200 people who matched the requirements and got the job.

- 45% had relevant experience. $P(\text{experience}) = 0.45$
- 60% had good grades. $P(\text{grades}) = 0.60$
- Of those with relevant experience, 50% also performed well academically. $P(\text{grades} \mid \text{experience}) = 0.50$

Question: what is the probability a candidate has relevant experience, given they had a high GPA?

$$P(\text{experience} \mid \text{grades}) = \frac{0.50 \times 0.45}{0.60} = \frac{0.225}{0.60} = 0.375$$

Compare the two directions:

$$\begin{aligned} P(\text{grades} \mid \text{experience}) &= 0.500 \\ P(\text{experience} \mid \text{grades}) &= 0.375 \end{aligned}$$

Since $0.5 > 0.375$, **it is more likely for an experienced candidate to also excel academically than for a high-grade student to have the required working background.** Candidates who did internships tend to also have solid GPAs.

The practical takeaway for the firm: hiring for experience is more likely to land the well-rounded candidate than hiring for grades.

Bayes with independent events

A useful sanity check. Take two events with nothing to do with each other: the weather being sunny, and your code compiling.

- $P(\text{code works}) = 0.3$
- $P(\text{sunny}) = 0.4$
- Since they are independent, $P(\text{sunny} \mid \text{code works}) = 0.4$ as well.

Plug into Bayes:

$$P(\text{code works} \mid \text{sunny}) = \frac{0.4 \times 0.3}{0.4} = \frac{0.12}{0.4} = 0.3$$

Unchanged. The chance of your algorithm performing as intended neither increases nor decreases based on the weather, and Bayes' theorem correctly reports that. You can blame the sun for bright screens, but not for your compiler errors.

13. From a Formula to a Way of Thinking

You have now used Bayes' theorem as a calculator. You were handed three numbers, you plugged them in, you got a fourth number out.

Bayesian inference is what happens when you stop treating this as arithmetic and start treating it as learning.

Here is the shift. Rename the pieces:

- Let **H** be a *hypothesis*, something you are unsure about. "This email is spam." "This patient has the disease." "This coin is biased."
- Let **E** be *evidence*, something you actually observed. "The email contains the word FREE." "The test came back positive." "The coin landed heads five times."

Now rewrite the identical formula:

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

Nothing changed mathematically. But read it out loud:

Given what I just saw, how much should I now believe my hypothesis?

That question is the whole subject. Bayesian inference is a procedure for **updating what you believe when new information arrives**.

14. The Four Ingredients

Every Bayesian calculation has exactly four parts. Learn their names, because textbooks and papers use them constantly.

Part	Symbol	Plain meaning
Prior	$P(H)$	What you believed <i>before</i> seeing the evidence
Likelihood	$P(E H)$	If the hypothesis were true, how likely is this evidence?
Evidence (or marginal likelihood)	$P(E)$	How likely is this evidence overall, under any hypothesis?
Posterior	$P(H E)$	What you believe <i>after</i> seeing the evidence

So the formula reads:

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}$$

A note on each one

The prior is your starting point. It might come from published statistics, historical data, a previous experiment, or a reasoned estimate. It is not a guess pulled from nowhere, and it is also not sacred. It is simply where you begin.

The likelihood is often the easiest to obtain, because it usually comes from a controlled study or a known mechanism. A lab can tell you "this test detects the disease 99% of the time in patients who have it." That is a likelihood.

The evidence is the term students most often mishandle. It is not "the probability the evidence is true." It is the probability of observing that evidence *at all*, across every possible state of the world. You compute it with the law of total probability from Section 9:

$$P(E) = P(E|H) \times P(H) + P(E|\text{not } H) \times P(\text{not } H)$$

With more than two hypotheses, sum across all of them:

$$P(E) = \sum P(E|H_i) \times P(H_i)$$

The posterior is the answer, and it is also your new starting point for next time. Hold on to that. We return to it in Section 17.

15. Worked Example: The Medical Test

This is the classic example, and it is classic because the answer surprises almost everyone the first time.

Setup

- A disease affects **1%** of the population.
- The test correctly returns positive for **99%** of people who have it.
- The test incorrectly returns positive for **5%** of people who do not.

Question: a patient tests positive. What is the probability they actually have the disease?

Most people answer "about 99%." Let's find out.

Step 1: Name the pieces

- H = patient has the disease
- E = patient tests positive
- Prior: $P(H) = 0.01$, so $P(\text{not } H) = 0.99$
- Likelihood: $P(E|H) = 0.99$
- False positive rate: $P(E|\text{not } H) = 0.05$

Step 2: Compute the evidence

$$\begin{aligned}P(E) &= (0.99 \times 0.01) + (0.05 \times 0.99) \\&= 0.0099 + 0.0495 \\&= 0.0594\end{aligned}$$

Step 3: Apply Bayes

$$P(H|E) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.167$$

The answer is about **16.7%**.

Why this happens

Imagine 10,000 people.

- 100 have the disease. The test catches **99** of them.
- 9,900 do not. The test wrongly flags **495** of them.
- Total positives: $99 + 495 = \mathbf{594}$
- Genuinely sick among those: $99 / 594 \approx \mathbf{16.7\%}$

The false positives swamp the true positives, because the healthy group is so much larger. A small error rate applied to a huge population produces more mistakes than a large success rate applied to a tiny population.

This is the single most important lesson in the handout. When a condition is rare, a positive result from a good test still leaves substantial doubt. Ignoring the prior is called the **base rate fallacy**, and it is one of the most common errors in applied reasoning.

Notice also that the test was not useless. It moved your belief from 1% to 16.7%, a seventeenfold increase. It just did not move it to certainty.

16. Worked Example: A Spam Filter

Now something closer to computer science. This is roughly how a naive Bayes classifier works.

Setup

- 40% of incoming mail is spam. $P(S) = 0.4$
- The word "FREE" appears in 30% of spam. $P(F|S) = 0.3$
- The word "FREE" appears in 2% of legitimate mail. $P(F|\text{not } S) = 0.02$

Question: an email contains "FREE." How likely is it to be spam?

Step 1 — Evidence.

$$\begin{aligned}
 P(F) &= (0.3 \times 0.4) + (0.02 \times 0.6) \\
 &= 0.12 + 0.012 \\
 &= 0.132
 \end{aligned}$$

Step 2 — Posterior.

$$P(S|F) = \frac{0.3 \times 0.4}{0.132} = \frac{0.12}{0.132} \approx 0.909$$

About 90.9%. One word pushed the estimate from 40% to nearly 91%.

17. Sequential Updating

Here is the property that makes Bayesian inference genuinely powerful rather than just clever.

Today's posterior becomes tomorrow's prior.

You do not have to process all your evidence at once. You can take it one piece at a time, and the result is the same.

Continuing the spam example

The same email also contains the word "WINNER."

- $P(W|S) = 0.15$
- $P(W|\text{not } S) = 0.005$

Your new prior is the posterior you just computed: $P(S) = 0.909$, so $P(\text{not } S) = 0.091$.

$$\begin{aligned}
 P(W) &= (0.15 \times 0.909) + (0.005 \times 0.091) \\
 &= 0.13635 + 0.000455 \\
 &= 0.136805
 \end{aligned}$$

$$P(S|W) = \frac{0.13635}{0.136805} \approx 0.997$$

Now 99.7%. Two words, two updates, and the filter is confident.

This is why spam filters sharpen as they read more of an email, why medical diagnosis proceeds test by test, and why a machine learning model can be trained on data arriving in a stream rather than all at once.

One caution

Chaining updates this way assumes the pieces of evidence are **conditionally independent** given the hypothesis. In our example, we assumed that among spam emails, containing "FREE" tells you nothing extra about whether it contains "WINNER."

That assumption is usually false in reality. Spam containing one marketing word is more likely to contain another. This is exactly why the method is called **naive** Bayes. The assumption is naive, but the classifier works well anyway, which is one of the pleasant surprises of the field.

18. The Odds Form

There is a second way to write Bayes' rule that is often faster by hand and gives better intuition.

Define **odds** as the ratio of a probability to its complement:

$$\text{odds}(H) = P(H) / P(\text{not } H)$$

A probability of 0.2 is odds of $0.2/0.8 = 0.25$, usually written 1:4.

Now the rule becomes remarkably clean:

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}$$

where the **likelihood ratio** is:

$$\text{LR} = \frac{P(E|H)}{P(E|\text{not } H)}$$

Redoing the medical test in odds form

- Prior odds: $0.01 / 0.99 = 1/99$
- Likelihood ratio: $0.99 / 0.05 = 19.8$
- Posterior odds: $(1/99) \times 19.8 = 0.2$, which is 1:5

Convert back to probability:

$$P = \text{odds} / (1 + \text{odds}) = 0.2 / 1.2 \approx 0.167$$

Same answer, and no need to compute $P(E)$ at all.

Why this form is useful

The likelihood ratio is a clean measure of **how much a piece of evidence is worth**, entirely separate from what you believed beforehand.

- LR **greater than 1** → the evidence supports the hypothesis
- LR **equal to 1** → the evidence is worthless, it says nothing
- LR **less than 1** → the evidence argues against the hypothesis

An LR of 19.8 is strong evidence. But strong evidence applied to very long prior odds still leaves you short of certainty. The odds form makes that tension visible in a single line.

Connecting back to independence

Recall Section 12: if H and E are independent, then $P(E|H) = P(E|\text{not } H)$, so the likelihood ratio is exactly **1**. Multiply your prior odds by 1 and nothing moves. The two ways of expressing "this evidence taught me nothing" agree perfectly.

19. Three Mistakes to Avoid

Mistake 1: Confusing $P(H|E)$ with $P(E|H)$

These are different quantities and frequently very different numbers. In legal contexts this is called the **prosecutor's fallacy**.

"The probability of this DNA match occurring by chance is 1 in a million, therefore there is a one in a million chance the defendant is innocent."

That reasoning is wrong. The first number is $P(\text{match} | \text{innocent})$. The second is $P(\text{innocent} | \text{match})$. Converting between them requires the prior, which is the size of the pool of people who could plausibly have been at the scene. If that pool is large, the second number can be nowhere near the first.

You met this already in Section 9: 15/47 of women were vegetarian, but 15/44 of vegetarians were women. Same numerator, different question, different answer.

Mistake 2: Ignoring the base rate

Covered in Section 15. When the prior is small, even strong evidence may leave the posterior small. Always ask "how common is this thing to begin with?" before interpreting a test result.

Mistake 3: Treating the posterior as final

The posterior is not a verdict. It is your current best estimate given the evidence available so far. More evidence can and should move it again. Treating a Bayesian result as permanent defeats the purpose of the method.

20. A Word on the Bigger Picture

You may hear about "Bayesian statistics" versus "frequentist statistics" as though they were rival camps. A brief and fair framing:

- The **frequentist** view treats probability as a long-run frequency. Parameters are fixed but unknown quantities, and it makes no sense to assign them probabilities.
- The **Bayesian** view treats probability as a degree of belief. Parameters can have distributions, and those distributions get updated as data arrives.

Both approaches are mathematically sound and widely used. The main practical debate concerns priors: critics argue that choosing a prior injects subjectivity, while defenders argue that every analysis contains assumptions and Bayesian methods at least state theirs explicitly.

For this course, you do not need to take a side. What matters is that you can carry out the calculation correctly and interpret the result honestly.

Practice — Part III

Problem 7 — Manufacturing. Factory A produces 60% of a company's chips, with a 2% defect rate. Factory B produces the remaining 40%, with a 5% defect rate. A randomly selected chip is defective. What is the probability it came from Factory A?

Problem 8 — Servers. Requests are routed to three mirrors. M1 handles 50% of traffic and times out 1% of the time. M2 handles 30% and times out 4% of the time. M3 handles 20% and times out 10% of the time. A request timed out. Compute the probability it was handled by each mirror, and identify the most likely culprit.

Problem 9 — Two coins. You have two coins. One is fair. The other has heads on both sides. You pick one at random and flip it twice. Both flips come up heads. What is the probability you picked the double-headed coin?

Problem 10 — Surveillance. A facial recognition system is run against a crowd of 100,000 people, exactly one of whom is a wanted suspect. The system is 99% accurate in both directions: it correctly identifies the suspect 99% of the time, and it incorrectly flags an innocent person 1% of the time. The system raises an alert on someone. What is the probability that person is the suspect?

Problem 11 — Odds form. Your prior odds for a hypothesis are 1:4. A piece of evidence arrives with a likelihood ratio of 6. What is the posterior probability?

Answers — Part III

Problem 7.

$$\begin{aligned} P(\text{defective}) &= (0.6 \times 0.02) + (0.4 \times 0.05) = 0.012 + 0.020 = 0.032 \\ P(A \mid \text{defective}) &= 0.012 / 0.032 = 0.375 \end{aligned}$$

37.5%. Note that although Factory A makes most of the chips, it is *less* likely than Factory B to be the source of a given defect, because its quality is much better.

Problem 8.

$$\begin{aligned} P(\text{timeout}) &= (0.5 \times 0.01) + (0.3 \times 0.04) + (0.2 \times 0.10) \\ &= 0.005 + 0.012 + 0.020 = 0.037 \\ P(M1 \mid \text{timeout}) &= 0.005 / 0.037 \approx 0.135 \\ P(M2 \mid \text{timeout}) &= 0.012 / 0.037 \approx 0.324 \\ P(M3 \mid \text{timeout}) &= 0.020 / 0.037 \approx 0.541 \end{aligned}$$

M3 is the most likely culprit at about 54%, despite handling the least traffic. The three posteriors sum to 1, which is a good habit to check.

Problem 9.

$$\begin{aligned}
 P(HH \mid \text{fair}) &= 0.25 & P(HH \mid \text{double}) &= 1 \\
 P(HH) &= (0.5 \times 0.25) + (0.5 \times 1) = 0.125 + 0.5 = 0.625 \\
 P(\text{double} \mid HH) &= 0.5 / 0.625 = 0.8
 \end{aligned}$$

80%. Two heads is decent evidence, but not conclusive. A fair coin does this one time in four.

Problem 10.

$$\begin{aligned}
 \text{Prior} &= 1 / 100,000 = 0.00001 \\
 P(\text{alert}) &= (0.99 \times 0.00001) + (0.01 \times 0.99999) \\
 &= 0.0000099 + 0.0099999 \approx 0.01001 \\
 P(\text{suspect} \mid \text{alert}) &= 0.0000099 / 0.01001 \approx 0.00099
 \end{aligned}$$

About 0.1%. Roughly 1,000 innocent people get flagged for every one genuine hit. This is the base rate fallacy at industrial scale, and it is a real and heavily debated problem in deployed surveillance systems.

Problem 11.

$$\begin{aligned}
 \text{Posterior odds} &= 1:4 \times 6 = 6:4 = 3:2 \\
 P &= 3 / (3 + 2) = 0.6
 \end{aligned}$$

60%.

Quick Reference

Set notation

$x \in A$	x is an element of A
$x \notin A$	x is not an element of A
$\forall$	for all
:	such that
$A \subseteq B$	A is a subset of B
$\emptyset$	the empty set

Combining events

$A \cap B$	intersection: both A and B
$A \cup B$	union: either A or B
$A \cup B = A + B - (A \cap B)$	
mutually exclusive $\Leftrightarrow A \cap B = \emptyset$	

Probability rules

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{requires } P(B) > 0$$

$$P(A \cap B) = P(A|B) \times P(B) \quad \text{multiplication rule}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \text{additive law}$$

$$P(A) = \sum P(A|B_i) \times P(B_i) \quad \text{law of total probability}$$

$$\text{independent} \Leftrightarrow P(A) = P(A|B) \Leftrightarrow P(A \cap B) = P(A) \times P(B)$$

Bayes

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

$$P(E) = \sum P(E|H_i) \times P(H_i)$$

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}$$

$$\text{odds} \rightarrow \text{probability: } P = \text{odds} / (1 + \text{odds})$$

The one-sentence summary of Part III: Start with what you already believed, multiply by how well the new evidence fits, normalize, and that is your updated belief.